

ALPENA COUNTY REGIONAL, MI, USA

WMO: 726390

Lat: 45.072N

Lon: 83.564W

Elev: 209

StdP: 98.85

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 94849

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.9	-17.8	-24.2	0.4	-20.1	-21.4	0.6	-17.2	9.6	-6.3	8.9	-5.8	1.7	270	0.584

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.3	31.1	21.7	29.2	20.8	27.5	19.9	23.4	28.6	22.3	27.1	21.2	25.5	4.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.6	16.7	26.0	20.6	15.7	24.9	19.7	14.8	23.7	70.4	28.9	66.2	27.2	62.5	25.6	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.9	8.0	7.3	-25.7	34.3	4.0	1.8	-28.6	35.5	-30.9	36.6	-33.2	37.6	-36.1	38.8		
			-25.9	25.4	3.9	1.3	-28.8	26.4	-31.1	27.2	-33.3	27.9	-36.1	28.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	6.9	-6.9	-6.5	-1.6	5.0	11.7	17.3	20.0	19.1	15.2	8.7	2.4	-3.1	(6)
(7)		DBStd	10.67	6.03	6.00	6.03	4.79	4.70	4.03	3.28	3.24	4.24	4.66	4.76	5.22	(7)
(8)		HDD10.0	2269	525	461	363	162	33	1	0	0	5	83	231	406	(8)
(9)		HDD18.3	4379	783	694	619	399	213	68	20	29	112	301	477	664	(9)
(10)		CDD10.0	1121	0	0	3	13	86	218	310	282	163	43	3	0	(10)
(11)		CDD18.3	191	0	0	0	1	8	36	72	53	19	2	0	0	(11)
(12)		CDH23.3	1893	0	0	3	14	123	407	689	454	179	23	0	0	(12)
(13)		CDH26.7	547	0	0	1	3	34	137	208	117	43	4	0	0	(13)
(14)	Wind	WSAvg	3.3	3.6	3.6	3.8	3.4	2.9	2.7	2.7	2.8	3.2	3.4	3.5	(14)	
(15)	Precipitation	PrecAvg	752	47	37	43	70	74	71	84	80	73	72	54	47	(15)
(16)		PrecMax	915	112	90	157	162	167	140	148	171	166	163	111	114	(16)
(17)		PrecMin	591	12	13	7	31	28	6	28	33	17	19	20	8	(17)
(18)		PrecStd	86	24	19	33	31	33	33	31	39	35	33	25	26	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.3	11.1	20.7	25.4	30.2	33.0	33.2	32.4	30.6	26.5	18.4	12.4	(19)
(20)		0.4%	MCWB	6.2	7.8	13.4	15.4	19.2	23.1	23.1	22.6	21.7	18.4	12.8	9.9	(20)
(21)		2%	DB	5.3	6.2	13.7	20.1	27.0	30.3	31.0	29.5	27.7	22.3	14.7	8.2	(21)
(22)		2%	MCWB	3.5	3.5	9.4	12.3	17.9	21.0	22.1	21.4	20.1	16.7	11.0	6.1	(22)
(23)		5%	DB	2.8	3.4	9.8	16.4	23.6	27.9	29.0	27.7	25.3	19.3	12.0	5.5	(23)
(24)		5%	MCWB	1.7	1.4	5.7	10.4	15.9	20.0	21.0	20.5	18.9	15.2	9.0	3.9	(24)
(25)		10%	DB	1.3	1.7	6.5	13.3	20.7	25.3	27.1	25.9	23.1	16.6	9.7	3.6	(25)
(26)		10%	MCWB	0.5	0.3	3.3	7.9	14.4	18.3	19.9	19.4	17.9	13.1	7.3	2.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.9	8.0	14.8	16.7	21.3	24.2	25.2	24.2	23.0	20.4	14.3	10.7	(27)
(28)		0.4%	MCDWB	7.8	10.2	19.4	22.5	28.0	31.3	31.1	29.0	28.4	23.8	17.1	11.9	(28)
(29)		2%	WB	3.6	3.8	10.1	13.6	19.2	22.4	23.6	22.8	21.3	17.6	11.7	6.5	(29)
(30)		2%	MCDWB	4.8	5.6	13.6	19.2	24.7	28.4	28.6	27.6	25.4	21.1	14.2	8.0	(30)
(31)		5%	WB	1.8	1.7	6.2	10.8	17.2	21.0	22.4	21.7	20.1	15.5	9.4	4.1	(31)
(32)		5%	MCDWB	2.7	3.1	9.3	15.3	22.4	25.9	27.0	25.8	23.8	18.9	11.8	5.3	(32)
(33)		10%	WB	0.5	0.5	3.6	8.3	15.1	19.6	21.2	20.6	18.8	13.6	7.6	2.3	(33)
(34)		10%	MCDWB	1.4	1.7	6.1	12.8	19.8	23.8	25.4	24.1	22.3	16.2	9.5	3.6	(34)
(35)	Mean Daily Temperature Range	MDBR	8.0	9.5	10.1	11.1	12.4	12.6	12.3	11.9	11.9	10.1	7.8	6.9	(35)	
(36)		5% DB	MCDBR	8.0	10.2	13.9	17.0	16.8	15.7	14.6	13.8	14.8	14.6	11.7	9.1	(36)
(37)		5% DB	MCWBR	7.0	8.1	9.4	10.5	9.0	8.0	7.1	7.0	8.1	9.4	8.9	7.5	(37)
(38)		5% WB	MCDBR	7.6	9.7	13.2	15.5	15.1	13.8	12.4	12.1	12.3	13.3	11.0	8.7	(38)
(39)		5% WB	MCWBR	6.8	8.0	9.6	10.5	8.9	7.8	6.9	6.8	7.5	9.2	8.9	7.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.285	0.297	0.308	0.350	0.371	0.403	0.402	0.390	0.368	0.338	0.307	0.280	(40)	
(41)		taud	2.415	2.355	2.444	2.371	2.386	2.318	2.341	2.370	2.427	2.492	2.524	2.480	(41)	
(42)		Ebn at Noon	835	889	931	911	898	866	862	860	852	833	806	808	(42)	
(43)		Edh at Noon	72	92	97	114	116	125	121	114	99	81	66	61	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.25	2.17	3.68	4.64	5.52	6.03	6.09	5.17	4.04	2.42	1.34	0.92	(44)	
(45)		RadStd	0.11	0.24	0.36	0.42	0.50	0.44	0.23	0.24	0.37	0.26	0.18	0.09	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (40 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page